

Demonstration of EarPressure VR: Ear Canal Pressure Feedback for Enhancing Environmental Presence in Virtual Reality

Seongjun Kang
HCIS Lab, GIST
Gwangju, Republic of Korea
ksjryan0728@gm.gist.ac.kr

Gwangbin Kim
HCIS Lab, GIST
Gwangju, Republic of Korea
gwangbin@gm.gist.ac.kr

Bocheon Gim
HCIS Lab, GIST
Gwangju, Republic of Korea
gimkenny1999@gm.gist.ac.kr

Jeongju Park
HCIS Lab, GIST
Gwangju, Republic of Korea
jeongjupark@gm.gist.ac.kr

Semoo Shin
HCIS Lab, GIST
Gwangju, Republic of Korea
sms081700@gm.gist.ac.kr

SeungJun Kim*
HCIS Lab, GIST
Gwangju, Republic of Korea
seungjun@gist.ac.kr



Figure 1: The EarPressure VR system simulates ambient pressure changes in virtual reality by modulating ear canal pressure between -40hPa and $+40\text{hPa}$ using a stepper motor-actuated syringe. Sealed earbuds enable both inward and outward pressure modulation, allowing the system to replicate sensations such as ear-plugged fullness during underwater descent.

Abstract

We demonstrate EarPressure VR, a novel haptic system that enhances environmental presence in virtual reality by simulating atmospheric pressure changes through controlled air pressure modulation in the user's ear canal. The system uses a VR headset with sealed earbuds and a motor-driven syringe to safely generate pressure variations that simulate sensations like underwater depth or high-altitude environments (within $\pm 40\text{hPa}$ of the ambient level). Attendees will experience two interactive scenarios: a gradual underwater descent, feeling a steady increase in inward pressure synchronized with depth; and instantaneous portal travel between high- and low-pressure worlds, feeling sudden and distinct environmental shifts. This demonstration showcases how leveraging the

ear canal as a haptic channel can create more physically grounded and immersive virtual experiences.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *Haptic devices*; *Virtual reality*.

Keywords

Ear Canal Pressure, Environmental Perception, Haptic Feedback, Virtual Reality

ACM Reference Format:

Seongjun Kang, Gwangbin Kim, Bocheon Gim, Jeongju Park, Semoo Shin, and SeungJun Kim. 2025. Demonstration of EarPressure VR: Ear Canal Pressure Feedback for Enhancing Environmental Presence in Virtual Reality. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25)*, September 28–October 01, 2025, Busan, Republic of Korea. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746058.3758999>

1 Introduction

Haptic feedback in virtual reality enhances the sense of presence by providing users with physical sensations [7]. Previous studies have primarily focused on object-level sensations, such as grasping virtual objects [20], colliding with obstacles [10, 15], or touching

*Corresponding author.

hot or cold surfaces [13, 16]. However, conveying ambient environmental conditions like atmospheric pressure changes remains largely unexplored. While VR systems have replicated temperature and humidity through thermal elements [12, 17] and mist generators [4, 9, 22], simulating atmospheric pressure presents unique challenges since users remain physically grounded in room-scale VR setups, whereas atmospheric pressure changes typically occur through substantial vertical movement [2, 3] or spatial displacement [6].

Human ears offer a promising solution for atmospheric pressure simulation. The middle ear is sensitive to subtle changes in ambient pressure [5], making it one of the earliest perceptible cues during ambient pressure variation [21]. The sensation of ear pressure can be induced by creating a pressure differential between the middle ear and the external ear canal [1], commonly experienced as ear fullness during activities like rapid altitude changes or diving.

EarPressure VR leverages this sensitivity by modulating air pressure within the user's ear canal using a sealed earbud system connected to a linear stepper motor-driven syringe (Figure 1). Inspired by tympanometry, a non-invasive diagnostic technique used in clinical settings to assess middle ear function [18], our system operates within safe clinical limits (± 40 hPa relative to ambient pressure), staying well below the pain-inducing threshold [8, 14] and standard tympanometry ranges [19].

This demonstration presents two interactive VR scenarios that showcase how ear pressure feedback enhances environmental presence: a gradual underwater descent simulation and instantaneous portal travel between different atmospheric conditions. Attendees will experience how this novel haptic modality creates more physically grounded and immersive virtual environments by providing ear canal feedback as atmospheric cues.

2 System Implementation

EarPressure VR is a self-contained, headset-mounted system designed to deliver synchronized pressure feedback to both ears. The system consists of two primary components: the pressure modulation units and a custom sealing interface. A detailed hardware specification and user study results can be found in Kang et al [11].

Hardware and Control. As shown in Figure 2, each side of the HMD is equipped with a Pressure Modulation Unit. This unit uses a linear stepper motor to drive a 3mL syringe, which precisely compresses or decompresses air in a sealed tube connected to the earbud. An integrated differential pressure sensor (HX710B) provides real-time feedback, enabling a closed-loop PID controller on an Arduino microcontroller to accurately maintain the target pressure. The entire system operates wirelessly within a safe range of ± 40 hPa relative to ambient pressure.

Sealing Interface. A reliable seal is critical for pressure modulation. We designed a custom four-flanged silicone ear tip that creates an airtight seal in the ear canal, preventing leakage during both inward (positive) and outward (negative) pressure changes.

3 Demonstration Experience

Attendees will wear a VR headset integrated with the EarPressure VR system. The 3-4 minute demonstration guides them through two

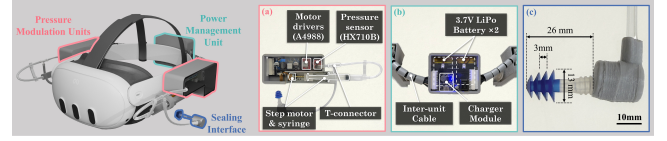


Figure 2: EarPressure VR hardware settings. (a) pressure modulation units. (b) power management unit. (c) Sealing plug.

interactive scenarios, each designed to showcase the system's capabilities in rendering both gradual and discrete pressure changes.

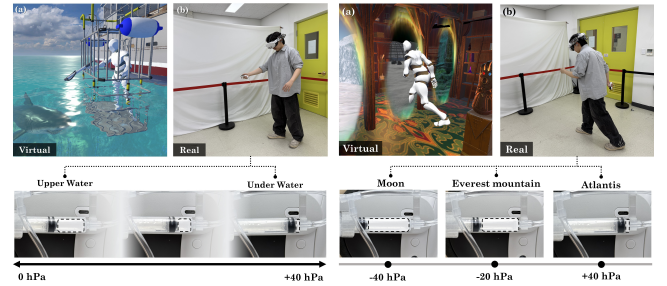


Figure 3: (a) The Jaws Cage Survival scenario simulates a gradual underwater descent. (b) The Sorcerer's Portal Room scenario demonstrates discrete environmental shifts.

3.1 Scenario 1: Jaws Cage Survival

In this scenario, attendees will find themselves in a shark cage. Using a controller, they can lower the cage deeper into the ocean.

Gradual Pressure Change: As the cage descends, attendees will feel a slow, steady increase in inward ear pressure, simulating the sensation of increasing water depth. The pressure is synchronized with their vertical movement, reaching a maximum of +40 hPa at the lowest point. Ascending back to the surface reverses the effect, with the pressure gradually returning to neutral.

3.2 Scenario 2: Sorcerer's Portal Room

Attendees are then transported to a magical room that contains portals to different worlds with distinct atmospheric conditions.

Discrete Pressure Change: Stepping through a portal to a high-altitude mountain peak instantly triggers an outward pressure (-20 hPa), simulating the thin atmosphere. A different portal leading to a deep-sea city induces an immediate inward pressure (+40 hPa). This allows attendees to feel the instant change between environments.

4 Conclusion

This demonstration introduces ear canal pressure as a novel haptic modality for VR. EarPressure VR enhances environmental presence by simulating virtual changes in altitude or depth. By showcasing continuous and discrete examples of inward and outward pressure, our system presents a practical approach for creating more physically grounded atmospheric sensations.

Acknowledgments

This work was partly supported by the Institute of Information & Communications Technology Planning & Evaluation (IITP)-ITRC (Information Technology Research Center) grant funded by the Korea government (MSIT) (IITP-2025-RS-2024-00436398, 33%), the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (RS-2024-00343397, 33%), and the Gwangju Institute of Science and Technology (GIST)–Massachusetts Institute of Technology (MIT) Research Collaboration Grant in 2025 (33%).

References

- [1] Bernard Ars and Joris Dirckx. 2016. Eustachian tube function. *Otolaryngologic clinics of north america* 49, 5 (2016), 1121–1133.
- [2] Kira Bacal, George Beck, and Michael R Barratt. 2008. Hypoxia, hypercarbia, and atmospheric control. *Principles of clinical medicine for space flight* (2008), 445–473.
- [3] Gerardo Bosco, Alex Rizzato, Richard E Moon, and Enrico M Camporesi. 2018. Environmental physiology and diving medicine. *Frontiers in psychology* 9 (2018), 72.
- [4] Yang-Sheng Chen, Ping-Hsuan Han, Kong-Chang Lee, Chiao-En Hsieh, Jui-Chun Hsiao, Che-Ju Hsu, Kuan-Wen Chen, Chien-Hsing Chou, and Yi-Ping Hung. 2018. Lotus: enhancing the immersive experience in virtual environment with mist-based olfactory display. In *SIGGRAPH Asia 2018 Virtual & Augmented Reality*. 1–2.
- [5] Joris JJ Dirckx and Jacob Sadé. 2005. Middle ear pressure regulation: basic research and clinical observation. *Otology & Neurotology* 26, 2 (2005), 300–309.
- [6] Thomas Ehlrl, Rory Smith, and Stefan Kaczmarczyk. 2017. Key dynamic parameters that influence ride quality of passenger transportation systems. *Transportation Systems in Buildings* 1, 1 (2017), 1–10.
- [7] Dominik Gall and Marc Erich Latoschik. 2018. The effect of haptic prediction accuracy on presence. In *2018 IEEE conference on virtual reality and 3D user interfaces (VR)*. IEEE, 73–80.
- [8] Tiffany A Glazer and Steven A Telian. 2016. Otologic hazards related to scuba diving. *Sports Health* 8, 2 (2016), 140–144.
- [9] Ping-Hsuan Han, Yang-Sheng Chen, Chiao-En Hsieh, Hao-Cheng Wang, and Yi-Ping Hung. 2019. Hapmosphere: Simulating the weathers for walking around in immersive environment with haptics feedback. In *2019 IEEE World Haptics Conference (WHC)*. IEEE, 247–252.
- [10] Seokhyun Hwang, Seongjun Kang, Jeongseok Oh, Jeongju Park, Semoo Shin, Yiyue Luo, Joseph DelPreto, Sangbeom Lee, Kyoobin Lee, Wojciech Matusik, et al. 2025. TelePulse: Enhancing the Teleoperation Experience through Biomechanical Simulation-Based Electrical Muscle Stimulation in Virtual Reality. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–26.
- [11] Seongjun Kang, Gwangbin Kim, Bocheon Gim, Jeongju Park, Semoo Shin, and SeungJun Kim. 2025. EarPressure VR: Ear Canal Pressure Feedback for Enhancing Environmental Presence in Virtual Reality. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology* (Busan, Republic of Korea) (UIST '25). Association for Computing Machinery, New York, NY, USA, Article 1, 16 pages. <https://doi.org/10.1145/3746059.3747618>
- [12] Seongjun Kang, Gwangbin Kim, Seokhyun Hwang, Jeongju Park, Ahmed Elsharkawy, and SeungJun Kim. 2024. Dual-sided Peltier Elements for Rapid Thermal Feedback in Wearables. *arXiv preprint arXiv:2405.11807* (2024).
- [13] Seongjun Kang, Gwangbin Kim, Seokhyun Hwang, Jeongju Park, Ahmed Ibrahim Ahmed Mohamed Elsharkawy, and SeungJun Kim. 2024. Flip-Pelt: Motor-Driven Peltier Elements for Rapid Thermal Stimulation and Congruent Pressure Feedback in Virtual Reality. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–15.
- [14] Stephen Chad Kanick and William J Doyle. 2005. Barotrauma during air travel: predictions of a mathematical model. *Journal of Applied Physiology* 98, 5 (2005), 1592–1602.
- [15] Pedro Lopes, Sijing You, Lung-Pan Cheng, Sebastian Marwecki, and Patrick Baudisch. 2017. Providing haptics to walls & heavy objects in virtual reality by means of electrical muscle stimulation. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 1471–1482.
- [16] Jeongju Park, Semoo Shin, Seongjun Kang, Gwangbin Kim, and SeungJun Kim. 2025. Magneto: Enabling Multimodal Haptic Feedback on Paper through Magnetic Fields. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–8.
- [17] Roshan Lalintha Peiris, Wei Peng, Zikun Chen, Liwei Chan, and Kouta Minamizawa. 2017. Thermovr: Exploring integrated thermal haptic feedback with head mounted displays. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 5452–5456.
- [18] Janet Shanks and Jack Shohet. 2009. Tympanometry in clinical practice. *Handbook of clinical audiology* 6 (2009), 159.
- [19] Janet E Shanks and David J Lilly. 1981. An evaluation of tympanometric estimates of ear canal volume. *Journal of Speech, Language, and Hearing Research* 24, 4 (1981), 557–566.
- [20] Shan-Yuan Teng, Tzu-Sheng Kuo, Chi Wang, Chi-huan Chiang, Da-Yuan Huang, Liwei Chan, and Bing-Yu Chen. 2018. Pupop: Pop-up prop on palm for virtual reality. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology*. 5–17.
- [21] R Thornton and P Vyrnwy-Jones. 1984. Environmental factors in helicopter operations. *BMJ Military Health* 130, 3 (1984), 157.
- [22] Fangtao Zhao, Ziming Li, Yiming Luo, Yue Li, and Hai-Ning Liang. 2024. AirWhisper: enhancing virtual reality experience via visual-airflow multimodal feedback. *Journal on Multimodal User Interfaces* (2024), 1–16.